



What is it?

Analytic Shape

Principle

Straight Lines

Circles

General Curves

Non-Analytic  
Shape

A Special Case

Generalized Hough  
Transforms (GHT)

GHT with Scaling and  
Rotation

# Chapter 7.1

## Hough Transforms

*Image Processing and Computer Vision*

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# What is it?

**Hough transforms** is a method for locating objects in input images.

## Questions

**Q1:** How do we specify objects being located?

**Q2:** Which information in the input image does Hough Transforms need?



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## Q1: How do we specify objects being located?

The objects can be expressed by one of the followings.

- ① **Analytic Form:** The objects are represented by mathematical relations, for examples,

$$\text{Straight line: } y = ax + b$$

$$\text{Circle: } (x - a)^2 + (y - b)^2 = r^2$$

$$\text{Ellipse: } \left( \frac{x - x_c}{a} \right)^2 + \left( \frac{y - y_c}{b} \right)^2 = 1$$

$$\text{General form: } f(\mathbf{x}, \mathbf{a}) = 0$$

- $\mathbf{x}, \mathbf{a}$  : vector of variables and parameters respectively.

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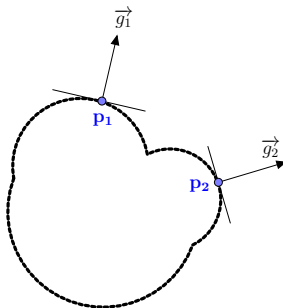
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# What is it?

- ② **Non-Analytic Form:** The objects are represented by the location and the gradient of pixels on the objects' boundary.



**Figure 1:** A shape represented by its boundary and gradients



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## Q2: Which information in the input image does Hough Transforms need?

Hough Transforms needs:

- 1 Edge pixels (location information)
- 2 Gradient of edge pixels (directional information)

⇒ First-order derivatives can be applied to obtain the required information.

# What is it?

A simple method for obtaining the required information from the input image:

## Example

- 1 Differentiate input image  $I(x, y)$  to obtain gradient image  $I_g(x, y)$ .
- 2 Find a threshold  $T$ , e.g.,  $T = \text{percentile } 90\% \text{ of } |I_g(x, y)|$
- 3 Obtain edge map:  $I_e(x, y) = |I_g(x, y)| > T$



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# Analytic Shape - Principle

Equation of straight lines:

**Image Space:** Treating  $a$  and  $b$  as constant parameters,  $x$  and  $y$  as variables

$$y = ax + b$$

**Parameter Space:** Treating  $x$  and  $y$  as constant parameters,  $a$  and  $b$  as variables

$$b = -xa + y$$



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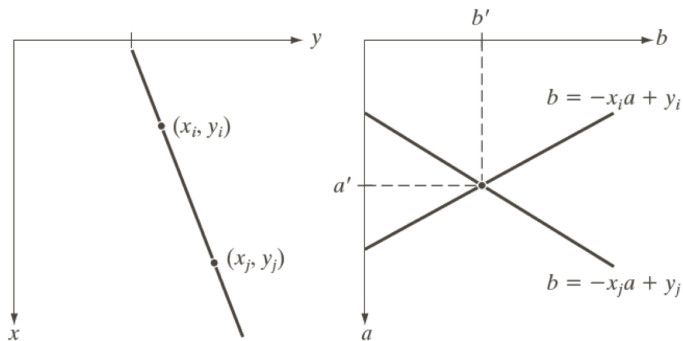
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# Analytic Shape - Principle



**Figure 2:** Left: **Image space**; Right: **Parameter space**

- **A point** in image space is **corresponding** to **a line** in parameter space.
- **A line** in image space is **corresponding** to **a point** in parameter space.



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If we have  $N$  edge points on the line passing two points, called  $(x_i, y_i)$  and  $(x_j, y_j)$  (see Fig. 2), then :

- We have  $N$  lines in parameter space
- These  $N$  lines intersect at **a common point**:  $(a', b')$  in parameter space

⇒ Detect this **common point** in parameter space ⇒  
equation of line in image space:  $y = a'x + b'$



So, the basic idea is:

- ① **Discretize** parameter space into small cells. Each cell contains the number of lines passing it
  - The whole space now called the **Accumulator**  $A(i, j)$
  - $i = 0, 1, \dots, M - 1$ ;  $j = 0, 1, \dots, N - 1$
- ② Find the common intersection point by finding the cell that contains the **largest number of lines** passing it. Assume that it is  $(a', b')$

The equation found is:  $y = a'x + b'$

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## Challenging Problems

- ① **Accuracy**: The accurate estimation of parameter  $a$  and  $b$  depends on the resolution of the accumulator, i.e., the size of cells in the accumulator
  - $\Rightarrow$  Discretize parameter space into smaller cells.
- ② **Memory cost**: The accumulator contains so many cells, especially, in the case that there are **many parameters** and that we use a **high resolution** accumulator.
- ③ **Large or unlimited range**: In some cases, parameters have large ranges, for example,  $a$  in  $y = ax + b$



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# Detection of Straight Lines



## Questions

- 1 In **line detection**, Parameter  $a$ , in  $y = ax + b$ , has an infinite range. How do we solve this problem?

## Equation of straight lines:

$$y = ax + b$$

Vertical line:  $a \rightarrow \infty$

$\Rightarrow$  use the following form

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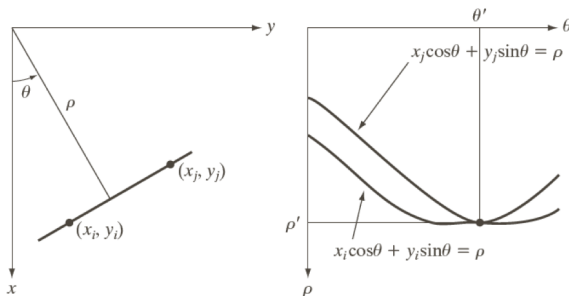
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# Analytic Shape - Detection of Straight lines



$$x \cos(\theta) + y \sin(\theta) = \rho$$



**Figure 3:** Hough transforms: (a) Image space, (b) Parameter space

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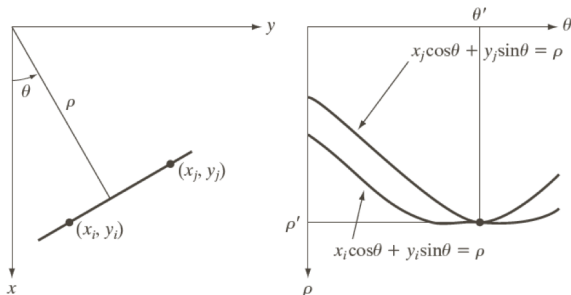
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# Analytic Shape - Detection of Straight lines



**Figure 4:** Hough transforms: (a) Image space, (b) Parameter space

- A fixed point in image space  $\Leftrightarrow$  A curve in parameter space
- A line in image space  $\Leftrightarrow$  A fixed point in parameter space



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## Advantages of the expression with $(\theta, \rho)$

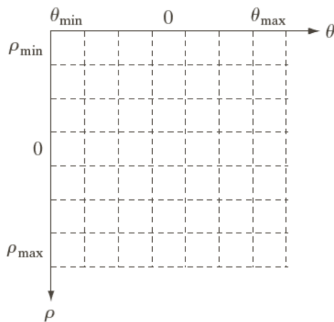
Range of  $\theta$  and  $\rho$  is **limited**.

- $-\pi \leq \theta \leq \pi$
- $-D \leq \rho \leq D$

$D$ : The maximum distance between two corners in images. Image's size:  $R \times C$ , then

$$D = \sqrt{R^2 + C^2}$$

# Analytic Shape - Detection of Straight lines



**Figure 5:** Demonstration of the discretization into  $M \times N$  cells

## Questions

- After discretization, how does cell's indices  $(i, j)$  relate to parameter  $\theta$  and  $\rho$ ?



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## Along $\rho$ -direction:

- $D = \sqrt{R^2 + C^2}$
- $\rho_{min} = -D; \rho_{max} = D$ : Left and right bound of the range
- $L_\rho = 2D$  : range's width
- $M$  : number of rows along  $\rho$  axis
- $i = 0, 1, \dots, M - 1$  : the index of cells along  $\rho$  axis
- $\Rightarrow$  Quantization step along  $\rho$  axis:  $\Delta_\rho = L_\rho / M$

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## Along $\rho$ -direction:

- $\Rightarrow$  From cell's index to  $\rho$  (at the center of the cell):

$$\begin{aligned}\rho &\stackrel{\Delta}{=} \text{cdm}_{idx2\rho}(i) \\ &= \rho_{min} + i \times \Delta_{\rho} + \frac{\Delta_{\rho}}{2}\end{aligned}$$

- $\Rightarrow$  From  $\rho$  to cell's index

$$\begin{aligned}i &\stackrel{\Delta}{=} \text{cdm}_{\rho2idx}(\rho) \\ &= \text{round} \left( \frac{\rho - \rho_{min}}{\Delta_{\rho}} \right)\end{aligned}$$

- ① **cdm**: continuous to **d**iscrete **m**apping
- ② **dcm**: **d**iscrete to **c**ontinuous **m**apping

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## Along $\theta$ -direction:

- $\theta_{min} = -\pi/2$ ;  $\rho_{max} = \pi/2$ : Left and right bound of the range
- $L_\theta = \pi$ : Range's width
- $N$  : number of columns along  $\theta$  axis
- $j = 0, 1, \dots, N - 1$  : the index of cells along  $\theta$  axis
- $\Rightarrow$  Quantization step along  $\theta$  axis:  
$$\Delta_\theta = L_\theta / N = \pi / N$$

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## Along $\theta$ -direction:

- $\Rightarrow$  From cell's index to  $\theta$  (at the center of the cell):

$$\begin{aligned}\theta &\triangleq \text{dcm}_{idx2\theta}(j) \\ &= \theta_{min} + j \times \Delta\theta + \frac{\Delta\theta}{2}\end{aligned}$$

- $\Rightarrow$  From  $\theta$  to cell's index

$$\begin{aligned}j &\triangleq \text{cdm}_{\theta2idx}(\theta) \\ &= \text{round}\left(\frac{\theta - \theta_{min}}{\Delta\theta}\right)\end{aligned}$$

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## Questions

③ How can we detect a straight line with Hough Transforms?



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## Algorithm: An Informal representation

### Algorithm 1 Hough Line Detection - PART1

- 1: Create an accumulator, referred to as  $A$
- 2: Set 0 for all cells in the accumulator
- 3: **for all** edge point in  $I_e(x, y)$  **do**
- 4:     **for all**  $j \in [0, N - 1]$  **do**  $\triangleright$  iterate on each cell along  $\theta$ -direction
- 5:          $\theta = \text{cdm}_{idx2\theta}(j)$
- 6:          $\rho = x \cos(\theta) + y \sin(\theta)$
- 7:          $i = \text{cdm}_{\rho2idx}(\rho)$
- 8:          $A(i, j) = A(i, j) + \Delta(x, y)$
- 9:     **end for**
- 10: **end for**





## Algorithm: An Informal representation

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### Algorithm 2 Hough Line Detection - PART2

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11: Find the largest value in the accumulator, assume at  $(s, t)$

12:  $\rho^* = \text{dcm}_{idx2\rho}(s)$

13:  $\theta^* = \text{dcm}_{idx2\theta}(t)$

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The detected line has following equation:

$$x\cos(\theta^*) + y\sin(\theta^*) = \rho^*$$

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## What is meaning of $\Delta(x, y)$ ?

- ①  $\Delta(x, y) = 1$  for any edge point
  - Accumulator  $A$  (with normalization) shows the **probability** of having a lines at each pair of  $(\rho, \theta)$
- ②  $\Delta(x, y) = |\vec{g}(x, y)|$ , where  $\vec{g}(x, y)$  is the gradient vector at edge point  $I_e(x, y)$ 
  - Accumulator  $A$  shows the strengthen of the **dis-continued information (edge)** along pixels on the straight line with parameter  $(\rho, \theta)$
- ③  $\Delta(x, y) = |\vec{g}(x, y)| + c$ , where  $c$  is a constant.
  - A variation from the previous

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## Exercise

- 1 Implement line detection with **Matlab** and **C/C++**
- 2 Assume that  $\phi(x, y)$  is the angle of the gradient vector at  $I_e(x, y)$  and that the estimation error of the gradient's angle is  $[-\Delta_\phi, +\Delta_\phi]$ . How does  $\phi(x, y)$  relate to parameter  $\theta$ ?
- 3 Using on  $\phi(x, y)$  and  $[-\Delta_\phi, +\Delta_\phi]$ , which cells in  $A$  should be increased for each  $\rho$ ?

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## Questions

⑤ How can we detect  $K$  straight lines with Hough Transforms in the input image?



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## Questions

- ⑤ How can we detect  $K$  straight lines with Hough Transforms in the input image?

## Guideline

- Create an accumulator, same as detecting 1 straight line.
- Use **non-maxima suppression** to remove (suppress) non-maxima cells.
- Find  $K$  **largest local maxima** by using **max-heap**



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## Algorithm 3 Hough Line Detection - PSEUDO-CODE

---

```
1: function DETECT_LINE(  
    REF  $I_e(x, y)$ : edge map,  
     $R, C$ : num of rows and cols of the edge map,  
     $M, N$ : num of rows and cols the accumulator  $A(i, j)$ ,  
    REF  $K$ : num of straight lines,  
    REF  $R_\theta, R_\rho$ : array of  $\theta$  and  $\rho$  detected )  
  
2:     Create Accumulator  $A$  with size  $M \times N$   
3:     COMP_ACCUMULATOR( $I_e, R, C, A, M, N$ );  
4:     APPLY_NONMAXIMA_SUPPRESSION( $A, M, N$ )  
5:     FIND_MAXIMA( $A, M, N, R_\theta, R_\rho, K$ )  
6: end function
```

---



---

## Algorithm 4 Updating the Accumulator, PART 1

---

```
1: function COMP_ACCUMULATOR(  
    REF  $I(x, y)$ : edge map,  
     $R, C$ : num of rows and cols of the edge map,  
    REF  $A$  : Accumulator  
     $M, N$ : num of rows and cols the accumulator  $A$  )  
  
2:   for  $r=0$  to  $M-1$  do  
3:     for  $c=0$  to  $N-1$  do  
4:        $A(r, c) = 0$ ;      ▷ Initialize the accumulator  
5:     end for  
6:   end for
```

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## Algorithm 5 Updating the Accumulator, PART 2

---

```
7:    $\Delta_{\theta} = L_{\theta}/N$ 
8:    $\Delta_{\rho} = L_{\rho}/M$ 
9:    $\rho_{min} = -D$  ;  $\theta_{min} = -\pi/2$ 
10:  for  $x = 0$  to  $R - 1$  do
11:    for  $y = 0$  to  $C - 1$  do
12:      if  $I_e(x, y) \neq 0$  then       $\triangleright (x, y)$ : edge point
13:        for  $j = 0$  to  $M - 1$  do
14:           $\theta = \text{dcm}_{idx2\theta}(j)$ 
15:           $\rho = x \cos(\theta) + y \sin(\theta)$ 
16:           $i = \text{cdm}_{\rho2idx}(\rho)$ 
17:           $A(i, j) = A(i, j) + \Delta(x, y)$        $\triangleright$  Voting
18:        end for
19:      end if
20:    end for
21:  end for
22: end function
```

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## Algorithm 6 Removing Non-maxima

---

```
1: function APPLY_NONMAXIMA_SUPPRESSION(  
    REF  $A$  : Accumulator  
     $M, N$ : num of rows and cols the accumulator  $A$  )  
  
2:   for all cell  $(i, j)$ , accept the border do  
3:      $NW = A(i - 1, j - 1)$   
4:      $N = A(i - 1, j)$   
5:      $NE = A(i - 1, j + 1)$   
6:      $E = A(i, j + 1)$   
7:      $SE = A(i + 1, j + 1)$   
8:      $S = A(i + 1, j)$   
9:      $SW = A(i + 1, j - 1)$   
10:     $W = A(i, j - 1)$   
11:     $C = A(i, j)$ 
```

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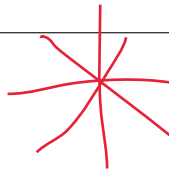
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## Algorithm 7 Non-maxima Suppression

---

```
12:       $C1 = (C < NW) \text{ OR } (C < N)$ 
13:       $C2 = (C < NE) \text{ OR } (C < E)$ 
14:       $C3 = (C < SE) \text{ OR } (C < S)$ 
15:       $C4 = (C < SW) \text{ OR } (C < W)$ 
16:      if ( $C1 \text{ OR } C2 \text{ OR } C3 \text{ OR } C4$ ) then
17:           $A(i, j) = 0$  ▷ suppress non-maxima
18:      end if
19:  end for
20: end function
```

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## Algorithm 8 Finding $K$ Maxima, PART1

---

```
1: function FIND_MAXIMA(  
    REF  $A$  : Accumulator  
     $M, N$ : num of rows and cols the accumulator  $A$   
    REF  $R_\theta, R_\rho$ : array of  $\theta$  and  $\rho$  detected  
    REF  $K$ : num of straight lines )  
  
2:     Create an empty max-heap, referred to as  $H_{max}$   
3:     for all cell  $(i, j)$  in  $A$ , accept the border do  
4:         if  $A(i, j) \neq 0$  then                                ▷ an extreme  
5:             key =  $A(i, j)$   
6:             data.rho =  $dcm_{idx2\rho}(i)$   
7:             data.theta =  $dcm_{idx2\theta}(j)$   
8:              $E = \{ \text{key, data.rho, data.theta} \}$   
9:             Add  $E$  to  $H_{max}$                                 ▷ re-heap up  
10:        end if  
11:    end for
```

---



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## Algorithm 9 Finding $K$ Maxima, PART2

---

```
12:    $k = 0$ 
13:   while ( $H_{max}$  is not empty) AND ( $k < K$ ) do
14:        $E =$  Remove maximum element from  $H_{max}$ 
15:        $R_{\rho}[k] = E.data.rho$ 
16:        $R_{\theta}[k] = E.data.theta$ 
17:        $k = k + 1$  ▷ next maximum
18:   end while
19:    $K = k;$  ▷ Update num of lines found
20: end function
```

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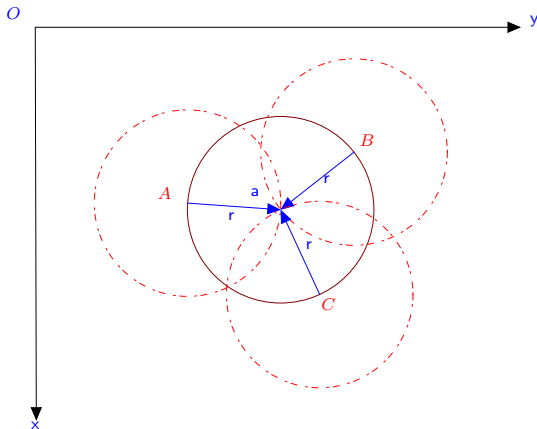
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# Detection of Circles

# Analytic Shape - Detection of Circles



**Figure 6:** A circle centered at  $a$ , radius  $r$ , contains three points  $A$ ,  $B$ , and  $C$ . We need to detect this circle.



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**Assumption:** We DO NOT know where is the center  $\mathbf{a}$ , but we know radius  $\mathbf{r}$  in advance.

## Facts

- $\mathbf{A}$  is on circle  $(\mathbf{a}, \mathbf{r}) \Rightarrow \mathbf{a}$  is on the circle centered at  $\mathbf{A}$ , radius  $\mathbf{r}$ . See Fig. 6



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**Assumption:** We DO NOT know where is the center  $\mathbf{a}$ , but we know radius  $\mathbf{r}$  in advance.

## Facts

- $\mathbf{A}$  is on circle  $(\mathbf{a}, \mathbf{r}) \Rightarrow \mathbf{a}$  is on the circle centered at  $\mathbf{A}$ , radius  $\mathbf{r}$ . See Fig. 6
- $\mathbf{B}$  is on circle  $(\mathbf{a}, \mathbf{r}) \Rightarrow \mathbf{a}$  is on the circle centered at  $\mathbf{B}$ , radius  $\mathbf{r}$ . See Fig. 6

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# Analytic Shape - Detection of Circles

**Assumption:** We DO NOT know where is the center  $\mathbf{a}$ , but we know radius  $r$  in advance.

## Facts

- $\mathbf{A}$  is on circle  $(\mathbf{a}, r) \Rightarrow \mathbf{a}$  is on the circle centered at  $\mathbf{A}$ , radius  $r$ . See Fig. 6
- $\mathbf{B}$  is on circle  $(\mathbf{a}, r) \Rightarrow \mathbf{a}$  is on the circle centered at  $\mathbf{B}$ , radius  $r$ . See Fig. 6
- $\mathbf{C}$  is on circle  $(\mathbf{a}, r) \Rightarrow \mathbf{a}$  is on the circle centered at  $\mathbf{C}$ , radius  $r$ . See Fig. 6
- **Center  $\mathbf{a}$  is the intersection of the three circles. See Fig. 6**

We can use the **voting-technique**, as used in line detection, to solve the detection



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## Circle equation:

### Explicit form

$$(x - x_c)^2 + (y - y_c)^2 = r^2$$

- Circle: has three parameters,  $x_c$ ,  $y_c$ , and  $r$
- $\Rightarrow$  Accumulator  $A$  is an array of 3 dimensions, indexed by  $x_c$ ,  $y_c$ , and  $r$
- $\Rightarrow A$  is a function of  $x_c$ ,  $y_c$ , and  $r$

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## Circle equation:

### Explicit form

$$(x - x_c)^2 + (y - y_c)^2 = r^2$$

- Range of  $x_c : 0, 1, \dots, R - 1$
- Range of  $y_c : 0, 1, \dots, C - 1$
- Range of  $r : 0, 2, \dots, R_{max} = \frac{\min(R, C)}{2}$
- Given an edge point  $(x_i, y_i)$  in image space:
  - For all points  $(x_c, y_c)$  in parameter space, compute dependent parameter  $r$  as follows:

$$r = \sqrt{(x_c - x_i)^2 + (y_c - y_i)^2}$$



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## Algorithm 10 Hough Circle Detection, Using explicit form

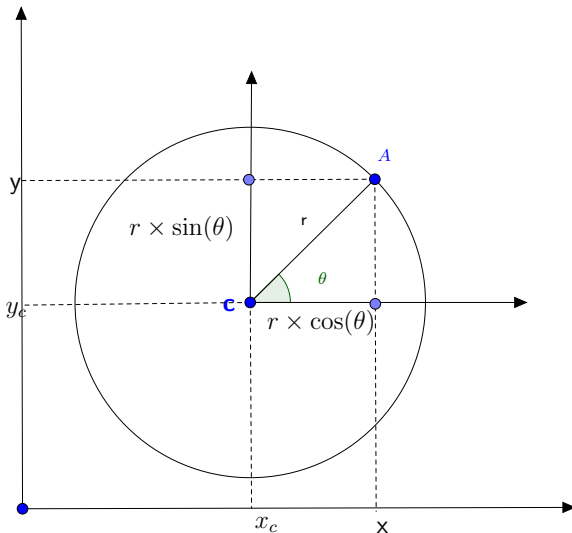
---

- 1: Create a 3D-Accumulator, referred to as  $A$
  - 2: Set 0 for all cells in the accumulator
  - 3: **for all** edge point in  $I_e(x, y)$  **do**
  - 4:     **for**  $x_c = 0$  to  $R - 1$  **do**
  - 5:         **for**  $y_c = 0$  to  $C - 1$  **do**
  - 6:             Compute  $r = \sqrt{(x_c - x)^2 + (y_c - y)^2}$
  - 7:              $A(x_c, y_c, r) = A(x_c, y_c, r) + 1$
  - 8:         **end for**
  - 9:     **end for**
  - 10: **end for**
  - 11: Find the largest cell in  $A(x_c, y_c, r)$ , assume at  $(x_c^*, y_c^*, r^*)$
- 

The detected circle has following equation:

$$(x - x_c^*)^2 + (y - y_c^*)^2 = r^{*2}$$

# Analytic Shape - Detection of Circles



**Figure 7:** Parametric form:  $\theta$  varies from  $0$  to  $2\pi \rightarrow \mathbf{A}$  draws a circle



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## Circle equation:

### Parametric form

$$x = x_c + r \cos(\theta)$$

$$y = y_c + r \sin(\theta)$$

- $\theta$  is **not** a **free** parameter
- Range of  $\theta$ :  $\theta \in [0, 2\pi]$

### Advantages of parametric form

Solve free parameters easily, for examples,

$$x_c = x - r \cos(\theta)$$

$$y_c = y - r \sin(\theta)$$



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## Algorithm: An Informal representation

### Algorithm 11 Hough Circle Detection - PART1

- 1: Create a 3D-Accumulator, named  $A$
- 2: Set 0 for all cells in the accumulator
- 3: **for all** edge point in  $I_e(x, y)$  **do**
- 4:     **for all**  $\theta_i \in [0, 2\pi]$  **do** ▷ Discretization of  $[0, 2\pi] \rightarrow \theta_i$
- 5:         **for all**  $r \in [r_{min}, r_{max}]$  **do**
- 6:              $x_c = x - r \cos(\theta)$
- 7:              $y_c = y - r \sin(\theta)$
- 8:              $i, j, k \leftarrow x_c, y_c$  and  $r$  respectively.
- 9:              $A(i, j, k) = A(i, j, k) + \Delta(x, y)$
- 10:         **end for**
- 11:     **end for**
- 12: **end for**



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## Algorithm: An Informal representation

---

### Algorithm 12 Hough Circle Detection - PART2

---

13: Find the largest cell in  $A(i, j, k)$ , assume at  $(i^*, j^*, k^*)$

14: Determine  $x_c^*, y_c^*$  and  $r^*$  from  $(i^*, j^*, k^*)$

---

The detected circle has following equation:

$$x = x_c^* + r^* \times \cos(\theta)$$

$$y = y_c^* + r^* \times \sin(\theta)$$

$$\theta \in [0, 2\pi]$$

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## Questions

⑦ How can we speed up the circle detection using gradient vectors at edge points?



## Questions

- ⑦ How can we speed up the circle detection using gradient vectors at edge points?

## Facts

- ① The direction of the gradient vector at every point  $(x, y)$  on a circle passes through the center of that circle.
- ② Angle  $\theta_i$  on Line 4 in Algorithm 11 and the angle of gradient vector at edge point  $I_e(x, y)$  on Line 3 must be coincided. See angle  $\theta$  in Figure 8

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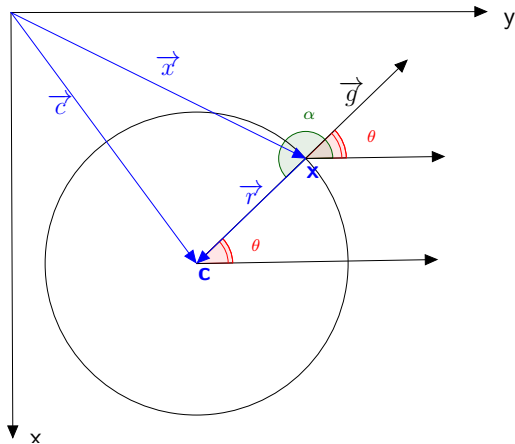
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**Figure 8:** Circle detection: relation between gradient vectors and the center



## Using gradient vectors

- $\theta_i$  in Line 4 (Algorithm 11) is equal to the angle of the gradient vector at edge point  $(x, y)$  on the circle.
- Let  $\phi(x, y)$  be gradient angle at edge point  $(x, y)$
- Let  $\Delta\phi$  be the maximum estimation error for gradient angle.
- Range of anticipated gradient angle:  
 $R_{grad} = [\phi(x, y) - \Delta\phi, \phi(x, y) + \Delta\phi]$
- So, Line 4 in previous algorithm will be changed to

**for all  $\theta_i \in [\phi(x, y) - \Delta\phi, \phi(x, y) + \Delta\phi]$  do**

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# Detection of General Curve



## General form of curves:

$$f(\mathbf{x}, \mathbf{a}) = 0$$

Where,

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_n \end{bmatrix}; \mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ \cdot \\ \cdot \\ \cdot \\ a_m \end{bmatrix}$$

- $n$  :  $n$  variables
- $m$  :  $m$  parameters

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## General form of circles:

$$\begin{aligned} f(\mathbf{x}, \mathbf{a}) &= 0 \\ \equiv (x - x_c)^2 + (y - y_c)^2 - r^2 &= 0 \end{aligned}$$

Where,

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}; \mathbf{a} = \begin{bmatrix} x_c \\ y_c \\ r \end{bmatrix}$$

- $n = 2$
- $m = 3$

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## Algorithm: An Informal representation

### Algorithm 13 Hough Curve Detection

- 1: Create accumulator  $A \equiv$  array of  $m$ -dimensions
- 2: Initialize  $A$  with 0 for all cells
- 3: **for all** edge point  $x_i$  **do**
- 4:     **for all** cell  $a_j$  **do**
- 5:         **if**  $f(x_i, a_j) == 0$  **then**
- 6:              $A(a_j) = A(a_j) + \Delta(x, y)$
- 7:         **end if**
- 8:     **end for**
- 9: **end for**
- 10: Find the largest cell in  $A$ , referred to as  $a^*$
- 11: **return**  $a^*$

The detected curve:  $f(\mathbf{x}, \mathbf{a}^*) = 0$





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# Detection of Non-Analytic Shapes



## Special Case: Detection of Circles

### Important Questions

- ① How does the **gradient direction** of a circle's edge point **relate** to the location of the **circle's center**?
- ② How can we generalize such the relationship for **more general shapes**?
- ③ How can we utilize **the generalized relationship** to detect a shape described by the shape's edge point?

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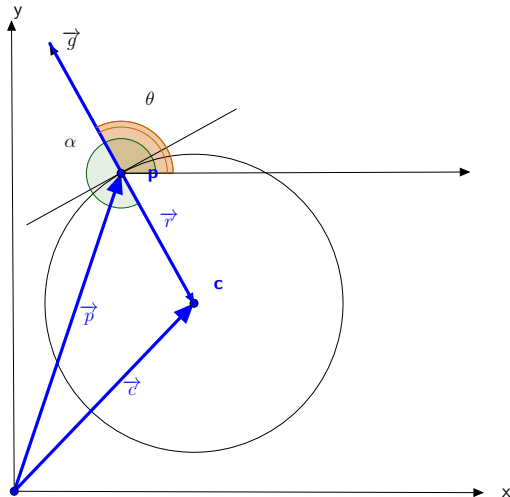
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# Non-Analytic Shape - Detection of Circles



**Figure 9: Circle:** Center  $\mathbf{c}$ , An edge point  $\mathbf{p}$ , Gradient vector at  $\mathbf{p}$ :  $\vec{g}$ , Angle of  $\vec{g}$ :  $\theta$



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## What we know

①  $\mathbf{p}$  or  $\vec{p}$ : is an edge point. Vector form of  $\mathbf{p}$  is  $\vec{p}$

$$\vec{p} = \begin{bmatrix} x \\ y \end{bmatrix}$$

②  $\theta$ : angle of gradient vector

③  $|\vec{r}|$ : radius of the circle being detected.

- $\vec{r}$  is the vector from  $\mathbf{p}$  to the center  $\mathbf{c}$  (not known now)
- We just know the magnitude of  $\vec{r}$

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## What we can infer

- ① Angle of  $\vec{r}$  :  $\alpha = \theta + \pi$
- ② Vector  $\vec{r}$ :

$$\begin{aligned}\vec{r} &= \begin{bmatrix} r \cos(\alpha) \\ r \sin(\alpha) \end{bmatrix} \\ &= \begin{bmatrix} r \cos(\theta + \pi) \\ r \sin(\theta + \pi) \end{bmatrix} \\ &= \begin{bmatrix} -r \cos(\theta) \\ -r \sin(\theta) \end{bmatrix}\end{aligned}$$

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Finally, the location of the center can be computed by:

$$\vec{c} = \vec{p} + \vec{r}$$

- Whenever we have  $\vec{r}$ , we know where the circle is.

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## Important Questions

- 1 How does the **gradient direction** of a circle's edge point **relate** to the location of the **circle's center**?

## Solution:

$$\vec{r'} = \begin{bmatrix} -r \cos(\theta) \\ -r \sin(\theta) \end{bmatrix}$$

- $\vec{r'}$  depends on the angle of gradient vector.
- $\vec{r'}$  **is a function of the angle of gradient vector.**

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## Important Questions

- 1 How does the **gradient direction** of a circle's edge point **relate** to the location of the **circle's center**?

## Solution:

$$\begin{aligned}\vec{c} &= \vec{p} + \vec{r} \\ &= \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} -r \cos(\theta) \\ -r \sin(\theta) \end{bmatrix} \\ &= \begin{bmatrix} x - r \cos(\theta) \\ y - r \sin(\theta) \end{bmatrix}\end{aligned}$$

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# Non-Analytic Shape - Generalized Hough Transforms (GHT)



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## Important Questions

- ② How can we generalize such the relationship for more general shapes?



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**SOLUTION: From circle to more general shapes**

**CIRCLE:**

- $|\vec{r}|$  is the same for every gradient vectors

**MORE GENERAL SHAPES:**

- $|\vec{r}|$  varies with the angle of gradient vector.



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**SOLUTION: From circle to more general shapes**

**CIRCLE:**

- $|\vec{r}|$  is the same for every gradient vectors
- Angle  $\alpha$  of  $\vec{r}$  is always  $(\theta + \pi)$ .

**MORE GENERAL SHAPES:**

- $|\vec{r}|$  varies with the angle of gradient vector.
- Angle  $\alpha$  of  $\vec{r}$  varies with the angle of gradient vector.



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**SOLUTION: From circle to more general shapes**

## CIRCLE:

- $|\vec{r}|$  is the same for every gradient vectors
- Angle  $\alpha$  of  $\vec{r}$  is always  $(\theta + \pi)$ .

## MORE GENERAL SHAPES:

- $|\vec{r}|$  varies with the angle of gradient vector.
- Angle  $\alpha$  of  $\vec{r}$  varies with the angle of gradient vector.
- One  $\theta$  can associated with more than one  $\vec{r}$

# Non-Analytic Shape - Generalized Hough Transforms



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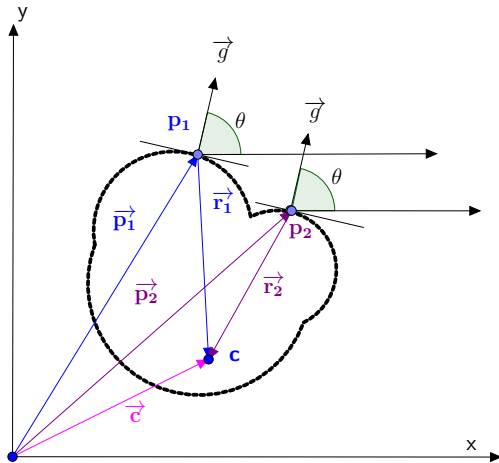
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**Figure 10:** Generalized Shape: An angle  $\theta$  can be associated with more than one vector  $\vec{r}$

# Non-Analytic Shape - Generalized Hough Transforms



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## Important Questions

- ③ How can we utilize the generalized relationship to detect a shape described by the shape's edge point?

# Non-Analytic Shape - Generalized Hough Transforms



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## Input

- ① An sample shape,  $S$ , described edge points on the shape boundary.
- ② An image contains shape  $S$

## Method for detecting generalized shapes

- ① **PHASE 1:** Describe the relationship between the gradient direction of edge points on  $S$  and a chosen point (referred to as **reference point**)  $c$  inside of  $S$ .
- ② **PHASE 2:** Detect instances of  $S$  in the input image.



## PHASE 1: Description of $\theta \rightarrow \vec{r}$

- ① Chose a point  $c$  inside of input shape  $S$ .
  - This point will be considered as the center of the shape, like center of a circle.
  - The purpose of **PHASE 2** is to detect  $c$
- ② Build an loop-up table that maps  $\theta$  (angle of gradient vectors) to  $\vec{r}$ 
  - Name of this mapping: **R-TABLE**
  - One  $\theta \rightarrow$  multiple  $\vec{r}$

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**Table 1:** R-Table illustration,  $R(\theta)$

| $i$   | $\theta_i$     | Content of the entry in R-Table                                  |
|-------|----------------|------------------------------------------------------------------|
| 0     | $\theta_0$     | $\vec{r}_{0,1}; \vec{r}_{0,2}; \dots; \vec{r}_{0,N_0}$           |
| 1     | $\theta_1$     | $\vec{r}_{1,1}; \vec{r}_{1,2}; \dots; \vec{r}_{1,N_1}$           |
| 2     | $\theta_2$     | $\vec{r}_{2,1}; \vec{r}_{2,2}; \dots; \vec{r}_{2,N_2}$           |
| ...   | ...            | ...                                                              |
| $M-1$ | $\theta_{M-1}$ | $\vec{r}_{M-1,1}; \vec{r}_{M-1,2}; \dots; \vec{r}_{M-1,N_{M-1}}$ |

- ① Number of entries:  $M$
- ②  $\theta_0$  : smallest angle of gradient vectors  $(-\pi)$
- ③  $\theta_{M-1}$  : largest angle of gradient vectors  $(+\pi)$
- ④  $N_i$ : number of vector  $\vec{r}$  associated with  $\theta_i$ 
  - $N_i$ : maybe a zero, maybe more than 1

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## Discretization of $\theta$

- $\theta_{min} = -\pi$
- $\theta_{max} = +\pi$
- Range of  $\theta$  :  $L_{\theta} = 2\pi$
- Number of table entries:  $M$
- $\Rightarrow \Delta_{\theta} = \frac{2\pi}{M}$



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From  $i$ , index of R-TABLE's rows, to  $\theta$  :

$$\begin{aligned}\theta &\triangleq \text{dcm}_{idx2\theta}(i) \\ &= \theta_{min} + i \times \Delta_{\theta} + \frac{\Delta_{\theta}}{2} \\ &= -\pi + i \times \Delta_{\theta} + \frac{\Delta_{\theta}}{2}\end{aligned}$$

From  $\theta$  to  $i$ , index of R-TABLE's rows:

$$\begin{aligned}i &\triangleq \text{cdm}_{\theta2idx}(\theta) \\ &= \text{round} \left( \frac{\theta - \theta_{min}}{\Delta_{\theta}} \right) \\ &= \text{round} \left( \frac{\theta + \pi}{\Delta_{\theta}} \right)\end{aligned}$$



## Input:

- $c$ : a chosen point in previous step.

---

### Algorithm 14 Generalized Hough Transforms: Building R-Table

---

- 1: Create **R-Table**  $R$  of  $M$  rows
  - 2: **for all** edge point  $p$  on shape  $S$  **do**
  - 3:     Compute vector  $\vec{r} = \vec{c} - \vec{p}$
  - 4:     Compute gradient vector  $\vec{g}$  at  $p$
  - 5:     Compute angle  $\theta$  of  $\vec{g}$
  - 6:     Determine row  $i = \text{cdm}_{\theta 2idx}(\theta)$
  - 7:     Add  $\vec{r}$  to Row  $i$  of  $R$
  - 8: **end for**
- 

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Rotation

## PHASE 2: Detecting instances of $S$

- 1 Create 2D-Accumulator  $A$  for each possible of center  $\mathbf{c}(x_c, y_c)$
- 2 Detect the largest cell in  $A$



What is it?

Analytic Shape

Principle

Straight Lines

Circles

General Curves

Non-Analytic  
Shape

A Special Case

Generalized Hough  
Transforms (GHT)

GHT with Scaling and  
Rotation

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## Algorithm 15 Detection of instances of $S$

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- 1: Create a 2D-Accumulator, referred to as  $A$
  - 2: Set 0 for all cells in the accumulator
  - 3: **for all** edge point in  $I_e(x, y)$  **do**
  - 4:     Create vector  $p = [x, -y]^T$      ▷ negative y, because  
      y-axis: upright, x-axis: to-right
  - 5:     Compute angle  $\theta$  of the gradient at  $I_e(x, y)$
  - 6:     Determine R-TABLE's row:  $l = \text{cdm}_{\theta 2id x}(\theta)$
  - 7:     Get List  $L$  of vector  $\vec{r}$  from Row  $l$
  - 8:     **for all** vector  $\vec{r}_i$  in  $L$  **do**
  - 9:         Compute vector  $\vec{c} = \vec{p} + \vec{r}_i$
  - 10:        Determine corresponding cell  $(x_c, y_c)$  in  $A$  from  $\vec{c}$
  - 11:         $A(x_c, y_c) = A(x_c, y_c) + \Delta(x, y)$
  - 12:     **end for**
  - 13: **end for**
  - 14: Find the largest cell in  $A(x_c, y_c)$ , assume at  $(x_c^*, y_c^*)$
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## Question

How can detect a shape in that case described as follows?

### Input:

- A sample of a shape  $S$  specified by the shape's edge points.
- An input image  $I(x, y)$

### Capability of the Detection:

- is able to detect instances  $S_i$  in the input  $I(x, y)$  in the case that  $S_i$  is a rotated and/or scaled version of  $S$  by an angle  $\alpha$  and a scaling factor  $s$

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## GHT without Scaling and Rotation

**R-Table:**  $R(\theta)$  is a multivalued vector function

- Input:  $\theta$ , for example,  $\theta = \theta_i$ , See Table 1
- Output: zero or multiple vectors:  $\vec{r}_{i,1}; \vec{r}_{i,2}; \dots; \vec{r}_{i,N_i}$

**Accumulator** :  $A(x_c, y_c)$  is a **2D-array**, indexed by the coordinates of the reference point  $c$





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What do Scaling and Rotation affect the shape  $S$ ?

## SCALING:

- Causes vector  $\vec{r}$  in R-Table scaled

## ROTATION:

- 1 Causes angle of gradient rotated an angle  $\alpha$
- 2 Causes vector  $\vec{r}$  in R-Table rotated an angle  $\alpha$

## ACCUMULATOR $A$ :

- 1 Need two more parameters: rotation angle  $\alpha$  and scaling factor  $s$
- 2  $\Rightarrow A(x_c, y_c, \alpha, s)$

## R-TABLE $R(\theta)$ :

- 1 Rebuilding is NOT required

## SCALING:

Scaling-matrix:

$$M_s = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$$

Scaled vector  $\vec{r}^s$  of  $\vec{r}$ :

$$\vec{r}^s = M_s \times \vec{r}$$

## Scaling and Accumulator

- Perform the scaling for all vectors in R-Table.
- Increase  $A$  for each scaling factor



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## ROTATION:

Rotation-matrix for rotation angle  $\alpha$ :

$$M_r = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{bmatrix}$$

Rotated vector  $\vec{r}^{rot}$  of  $\vec{r}$ :

$$\vec{r}^{rot} = M_r \times \vec{r}$$

## Rotation and Accumulator

- Perform the rotation for all vectors in R-Table.
- Increase  $A$  for each rotation angle  $\alpha$ ,  $0 \rightarrow 2\pi$  in general case.

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## Steps to rotating the whole shape $S$ by $\alpha$

- ① All R-Table's indices are increased by  $-\alpha$ , takes **modulo** by  $2\pi$  after increasing.
  - $\theta$  : angle of gradient vector.
  - Compute  $\theta^{rot} = (\theta - \alpha) \text{ modulo } 2\pi$
  - $\equiv$  Treat R-Table as a circular buffer, shift  $\theta$  around the circular buffer an amount  $-\alpha$
- ② All vectors found at  $\theta^r$  are rotated by  $\alpha$

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**Algorithm 16** Detection of instances of  $S$  with scaling and rotation

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- 1: Create a 4D-Accumulator, referred to as  $A(x_c, y_c, r, s)$
  - 2: Set 0 for all cells in the accumulator
  - 3: **for all** edge point in  $I_e(x, y)$  **do**
  - 4:     Create vector  $p = [x, -y]^T$      ▷ negative y, because  
      y-axis: upright, x-axis: to-right
  - 5:     Obtain gradient vector  $\vec{g}$  at  $I_e(x, y)$
  - 6:     Compute angle  $\theta$  of  $\vec{g}$
  - 7:     **for all** rotation angle  $\alpha$  **do**
  - 8:         Compute  $\theta^{rot} = (\theta - \alpha)$  **modulo**  $2\pi$
  - 9:         Find R-TABLE's row:  $l = \text{cdm}_{\theta 2idx}(\theta^{rot})$
  - 10:        Get List  $L$  of vector  $\vec{r}$  from Row  $l$
  - 11:        Compute rotation matrix  $M_r(\alpha)$
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**Algorithm 17** Detection of instances of  $S$  with scaling and rotation

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12:      for all scaling factor  $s$  do
13:          Compute scaling-matrix  $M_s(s)$ 
14:          for all vector  $\vec{r}_i$  in  $L$  do
15:              Transform  $\vec{r}_i^{trans} = M_r \times M_s \times \vec{r}_i$ 
16:              Compute  $\vec{c} = \vec{p} + \vec{r}_i^{trans}$ 
17:              Determine  $(x_c, y_c)$  from  $\vec{c}$ 
18:               $A(x_c, y_c, r, s) = A(x_c, y_c, r, s) + \Delta(x, y)$ 
19:          end for
20:      end for                                ▷ scaling factor  $s$ 
21:  end for                                    ▷ rotation angle  $\alpha$ 
22: end for                                    ▷ edge point  $I_e(x, y)$ 
23: Find the largest cell in  $A(x_c, y_c, r, s)$ , assume at
     $(x_c^*, y_c^*, r^*, s^*)$ 
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